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Gates, Bezos bet on flow battery technology, a potential rival to big bets on lithium-ion

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Tom Connor, special to CNBC.com

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KEY POINTS

The U.S. energy storage market is expected to grow by a factor of 12 in the next five years, from 430MW deployed in 2019 to more than 5GW and a value of more than \$5 billion by 2024, says Wood Mackenzie Energy Storage Service.

Tesla and GM are making big bets on lithium-ion batteries for energy storage systems and electric vehicles, but billionaire investors and venture capital firms are investing in competing battery technology, such as flow batteries.

Breakthrough Energy Ventures, the group of private investors led by Bill Gates and fellow billionaires Jeff Bezos, Michael Bloomberg, Richard Branson and Jack Ma, invested in iron-flow battery maker ESS in November.



ESS, which makes long-duration, iron flow batteries, secured \$30 million in a Series C investment round from Breakthrough Energy Ventures (BEV), the group of private investors led by Bill Gates and fellow billionaires Jeff Bezos, Michael Bloomberg, Richard Branson and Jack Ma, among others.

ESS

A UN report on climate change released Nov. 26 amounted to a dire warning for Earth: Unless greenhouse gas emissions are drastically reduced, and soon, the planet faces dangerously and irreversibly high temperatures in the near future. The report also criticized the 195 nations that signed the 2015 Paris Agreement for not doing nearly enough to reduce emissions. Two days earlier the World Meteorological Organization reported that greenhouse gases reached a record high in 2018, with no sign of peaking.

The warnings, albeit ominous, may prove timely for some investors.

In the wake of recent catastrophic storms in the Caribbean, along with devastating fires and mandatory power shutoffs in California, billionaire investors and venture capital firms are viewing renewable energy storage systems as a stable bet in an unstable future.

The U.S. energy storage market is expected to grow by a factor of 12 in the next five years — from 430 megawatts deployed in 2019 to more than 5 gigawatts —

storage market value in the U.S. alone will be \$5.5 billion by 2024.

Lithium-ion vs. iron-flow battery tech

Energy storage systems enable commercial enterprises and power-sensitive facilities, such as hospitals, to continue running when traditional power sources and generators fail or are unable to function. In addition, clean energy batteries have proved to be an environmentally safer, lower-cost alternative to carbon-based fuels. They also represent a sustainable way to deal with the intermittency of renewable energy from solar and wind.

In the early-1990s, lithium-ion energy storage systems replaced nickel cadmium batteries to serve the burgeoning cellphone and consumer electronics markets. More recently, they are being used in medical equipment and electric vehicles.

[Tesla](#) is building massive “gigafactories” to produce lithium-ion batteries for electric vehicles and Tesla Energy’s storage solutions business, including its newest Gigafactory 3 in Shanghai, China. [GM](#) just announced a multibillion-dollar investment in a lithium-ion battery plant in Ohio.



VIDEO 02:18

GM to invest up to \$2.3 billion in battery plant joint venture with LG Chem

charged and discharged, eventually needing replacement, and on occasion have exploded or caught fire. Iron low-energy storage systems, by contrast, last indefinitely, with no environmental risks. Both systems store energy from solar, wind and water on power grids, pulling it off as needed and re-injecting it when not.

Cost drives energy markets, and clean energy storage systems have gained momentum as technologies have advanced and manufacturing costs have dropped, particularly for lithium-ion batteries. From 2010 to 2018, lithium-ion battery costs declined by 85% and are forecast to decline by another 50% by 2030, according to [Bloomberg New Energy Finance](#), with major implications for the utility and electric vehicle market.

Gates, Bezos, Bloomberg money

Unlike other grid-scale storage technologies, lithium ion has benefited from the manufacturing scale of both consumer electronics and electric vehicles markets. Together the two industries have dominated the vast majority of lithium-ion markets, resulting in cost declines.

And with plunging system costs come the increasing value of clean energy storage, especially for new technologies that extend battery duration.

“Right now we’re only looking at energy storage that are around four hours,” says Daniel Finn-Foley, head of the energy storage team at Wood Mackenzie. “And lithium ion can still compete at four hours. But as we move into the next phase of the energy transition, four hours isn’t going to cut it anymore. That’s why all these investors are asking if there’s a longer-duration technology that scales better than lithium ion. This is where flow batteries come in.”

“You can follow the money,” Finn-Foley said. “All the investors are smelling it.”

In October of this year, ESS, a manufacturer of low-cost, long-duration, iron-flow batteries for the global renewable energy infrastructure, secured \$30 million in a Series C investment round from Breakthrough Energy Ventures, the group of private investors led by Bill Gates and fellow billionaires Jeff Bezos, Michael Bloomberg,

group issued a statement that said, “There is tremendous need for low maintenance storage technology that can extend the availability of intermittent renewable energy sources like wind and solar.”

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“We’ve seen more interest in energy storage, particularly with storms in the Caribbean,” said Craig Evans, co-founder and CEO of ESS, based in Wilsonville, Oregon. “There’s also talk of using our technology in California. But I think the reason for this investment is that the investors see our technology as one that offers consumers a number of benefits over lithium ion — longer duration in order to have solar-plus as a base load, the ability to work in very hot environments, and indefinite cycles at a similar price point.”

Flow battery technology is [not the only bet the big money is making](#) on competitors to traditional lithium-ion batteries. Tesla employee No. 7, Gene Berdichevsky, who worked on the original Tesla Roadster’s battery, is now CEO of Sila Nanotechnologies, which took in \$70 million from VC investors in 2018 for its silicon anode battery technology. Enevate, another silicon anode battery start-up — which is sometimes referred to as “next-gen” lithium ion — has raised more than \$100 million in VC funding and includes GM battery partner LG Chem among its recent investors.

As opportunity inherent in clean batteries turn to necessity, investors will increasingly look for technologies that can help make vulnerable regions of the world resilient in future energy crises. Part of this happening will depend on regulatory policy and the point at which the U.S. sets 100% clean energy goals.

In the future, according to Finn-Foley, utilities will have the opportunity to aggregate and distribute alternative energy assets from homes and businesses — solar demand

minimizing conventional generation through unconventional means.

“We’ll start to see residential business models all over the place,” he said. “For now, it’s still very hard to express the value that that resiliency provides, but the day is coming.”

—By Tom Connor, special to CNBC.com

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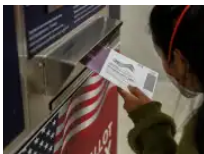
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